Inhomogeneous self-consistent mean-field theory of zinc doping of the cuprates

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1 Introduction

J.C. Seamus Davis discussed with me the problem of zinc doping in the cuprates. It unknown why the superconductivity is killed by small doping. Seamus suggested I try modelling the situation by switching off the Hubbard U on the zinc site. Copper peroxide

(CuO_2) is stabilised by surrounding elements in the cuprate, thought to be irrelevant to superconductivity due to the lack of supercurrent along the third axis. In addition to the stabilisers, the ingredients for a cuprate superconductivity are copper (Cu) whose d-wave orbital provides unconventional hopping, and oxygen (O_2) which staggers the hopping. The minimal model of superconductivity is onsite singlet pairing, and hence the d-wave pairing is irrelevant to the interaction term. More unconventional models of the superconductor may include multiorbital triplet pairing, in which the d-wave and p-wave symmetries would become relevant to the interaction (e.g. nearest-neighbour pairing in the bulk via Monte-Carlo [Kobayashi and Yokoyama REF)]. We consider the zinc-doped minimal model.

H. Alloul *et al* review Defects in correlated metals and superconductors [CITE:arXiv:0711.0877v1 (2018)].

2 Topology

Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)] showed topological modes when the interdimer hopping is different from the intradimer hopping, with equal chemical potential.

Have they checked with a rigid shift?

Diagonal hoppings within the $(\text{Cu-O})_4$ cluster

Any nonzero CDW-like density in the Fock channel would give rise to superconductivity.

3 Review of defects in correlated metals

H. Alloul *et al* [CITE:arXiv:0711.0877v1 (2018)] review theories of impurities in d-wave superconductor theories of impurities

4 Reproducing results

The normal state Hamiltonian describing the copper peroxide plane with zinc impurity doping is given by

$$\begin{aligned}\hat{\mathcal{H}}_0 \equiv & \sum_{-N/2\leq x,y\leq N/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\sum_{m\in\{d,p_x,p_y\}}\left[-\mu_m \hat{c}^{\sigma,m\dagger}_{x,y}\hat{c}^{\sigma,m}_{x,y}\right] + \sum_{-L/2\leq x,y\leq L/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left[+V\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,d}_{x,y}\right] \\ & + \sum_{-N/2\leq x,y\leq N/2}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left[+t\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,p_x}_{x,y} - t\hat{c}^{\sigma,d\dagger}_{x,y}\hat{c}^{\sigma,p_y}_{x,y} - t\hat{c}^{\sigma,p_x\dagger}_{x,y}\hat{c}^{\sigma,d}_{x+1,y} + t\hat{c}^{\sigma,p_y\dagger}_{x,y}\hat{c}^{\sigma,d}_{x,y+1}\right] \quad (1)\end{aligned}$$

where the fermion operator $\hat{c}^{\sigma,m\dagger}_{x,y}$ creates an operator on atomic orbital $m \in \{d_{x^2-y^2}, p_x, p_y\}$ within the unit cell located at $\mathbf{R}_{x,y} = x\mathbf{b}_0\hat{x} + y\mathbf{b}_1\hat{y}$, where $\mathbf{b}_0, \mathbf{b}_1$ are the lattice basis vectors, $\hat{n}^{(\alpha)\dagger}_{x,y}$ is the number operator, $\mu_d \equiv \mu$ and $\mu_{p_x} \equiv \mu_{p_y} \equiv \mu + s$ are the chemical potentials

of the copper and oxygen orbitals and s is a rigid shift of the atomic potentials. The parameter V sets the coupling strength of a zinc impurity at the centre of our lattice, and whose size goes from a single site $L = 0$ to a cluster $L > 0$. In the minimal theory, we assume half-filling $\mu = s = 0$ and zero coupling $V = 0$. The interorbital, intra- and inter-unit-cell hoppings strengths are given by $t = 1$, which normalises the model paramters. The boundary conditions are taken to be periodic. The lattice is depicted with $N = 3$ unit cells in Figure 1.

The interacting part of the minimal theory describes a singlet electron-electron interaction on the copper sites, but not on the zinc-doped sites

$$\hat{\mathcal{V}} \equiv \sum_{-N/2 \leq x,y \leq N/2} \sum_{m \in \{d,p_x,p_y\}} \left[+U_m \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m\dagger}_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \right], \tag{2}$$

where U_m is the strength of the Hubbard interaction on orbital m, which, in the simplest theory are $U_m \equiv U$. $U > 0$ describes Coulomb repulsion and $U < 0$ is the attractive BCS (singlet) scenario. The minimal model describing Zn-doped CuO_2 superconducting planes with BCS-like interactions on the Cu sites is then

$$\hat{\mathcal{H}} \equiv \hat{\mathcal{H}}_0 + \hat{\mathcal{V}} \tag{3}$$

We solve this Hamiltonian with inhomogeneous lattice self-consistent mean-field theory.

5 Self-consistent mean-field solution

The Bogoliubov-de Gennes (BdG) mean-field Hamiltonian is

$$\begin{aligned} \hat{\mathcal{V}}_{\mathrm{mf}} = & \sum_{\sigma \in \{\uparrow,\downarrow\}} \sum_{m \in \{d,p_x,p_y\}} \sum_{-N/2 \leq x,y \leq N/2} \left[\phi^{\sigma,m}_{x,y} \hat{n}^{\sigma,m}_{x,y} \right] \\ & + \sum_{m \in \{d,p_x,p_y\}} \sum_{-N/2 \leq x,y \leq N/2} \left[\Phi^{m}_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} + \Delta^{m}_{x,y} \hat{c}^{\uparrow,m\dagger}_{x,y} \hat{c}^{\downarrow,m\dagger}_{x,y} + \Delta^{m*}_{x,y} \hat{c}^{\uparrow,m}_{x,y} \hat{c}^{\downarrow,m}_{x,y} \right], \end{aligned} \tag{4}$$

where ϕ is the *Hartree* term which renormalises the chemical potential, allowing for a Peirl's instability. Φ is a *Fock* term, describing spin-flips on the d-orbitals. The additional *Gorkov* terms Δ_v capture fermion-fermion pairing, relevant to theories of magnetism ($U > 0$) and superconductivity ($U < 0$). These terms, solved self-consistently, are known as inhomogenous *(superconducting) density functional theory, SC-DFT*. The self-consistent solution to the mean field is the saddle point $\langle \hat{\mathcal{H}} \rangle_{\mathrm{mf}} = \langle \hat{\mathcal{H}}_{mf} \rangle_{\mathrm{mf}}$, in which the mean-field eigenstates $\{E_n, |n\rangle\}$ approximate the state of the interacting theory by minimising (or, unstably, maximising) the free energy, where $E_n \geq 0$ are half the solutions to the Bogoliubov-de Gennes (BdG) Hamiltonian, and where the rest of the BdG spectrum $E_n \leq 0$

are related by particle-hole symmetry. The solutions are

$$\phi_{x,y}^{\uparrow,m} = +U_m \langle \hat{n}_{x,y}^{\downarrow,m} \rangle, \tag{5}$$

$$\phi_{x,y}^{\downarrow,m} = +U_m \langle \hat{n}_{x,y}^{\uparrow,m} \rangle, \tag{6}$$

$$\Phi_{x,y}^{m} = -U_m \langle \hat{c}_{x,y}^{\uparrow,m\dagger} \hat{c}_{x,y}^{\downarrow,m} \rangle, \tag{7}$$

$$\Delta_{x,y}^{m} = +U_m \langle \hat{c}_{x,y}^{\uparrow,m} \hat{c}_{x,y}^{\downarrow,m} \rangle, \tag{8}$$

$$\Delta_{x,y}^{m*} = +U_m \langle \hat{c}_{x,y}^{\downarrow,m\dagger} \hat{c}_{x,y}^{\uparrow,m\dagger} \rangle. \tag{9}$$

Note the sign of the Fock terms due Fermionic anticommutation. In the simplest theory, we let $U_m \equiv U$.

6 Free energy

At zero temperature, the free energy of the interacting theory is

$$\mathcal{F} =, \tag{10}$$

where the mean-field free energy is

$$\mathcal{F}_{\rm mf} = E_g - 2k_B T \sum_n \left[\log 1 + e^{-E_n/k_B T} \right], \tag{11}$$

where T is the temperature of the thermal bath enclosing the system, k_B is the Boltzmann constant, E_n are the eigenvalues of the mean-field Bogoloiubov-de Gennes Hamiltonian $\mathcal{H}_{\rm mf}$ and the ground state energy is given by

$$E_g = \sum_n [E_n] + \sum_\sigma \sum_m \sum_{x,y} \left[\langle \hat{n}_{x,y}^{\sigma,m} \rangle \right] / \mathcal{N}, \tag{12}$$

and $\mathcal{N}$ is the number of degrees of freedom. The quartic interaction $\langle \mathcal{V} \rangle_{\rm mf}$ in the mean-field basis is given by

$$\langle \mathcal{V} \rangle_{\rm mf} =, \tag{13}$$

More details of the inhomogeous self-consistent superconducting density functional theory can be found in Leggett [REF].

7 Green's function

The solution to the Hamiltonian is given by the Green's function, which we remind the reader, is given by

$$\hat{\mathcal{G}}(\omega - i\epsilon, x, y) \equiv \sum_n \left[\frac{\psi_n(x,y)\psi_n(x,y)^\dagger}{\omega - i\epsilon - E_n} \right], \tag{14}$$

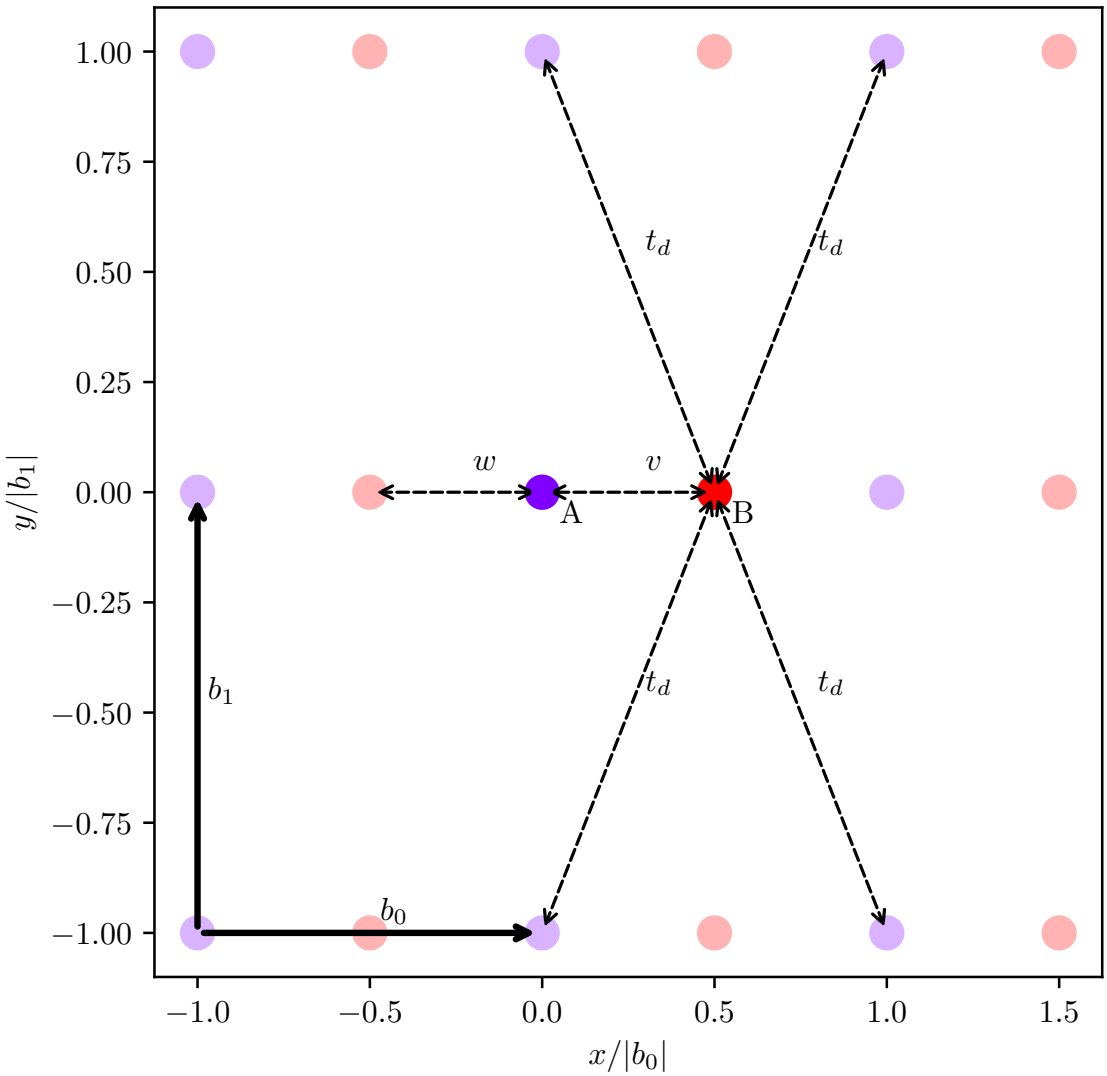

Figure 1: Two-dimensional multiorbital Weyl SSH model. Along the $\hat{x}$-direction, atomic sites A and B within the unit cell form an SSH chain, with intracell hopping v and intercell hopping w. An additional interchain hopping parameter couples the chains into a two-dimensional stack. Following Rosenberg and Manousakis, we study a diagonal hopping t_d, which is a Type II extension of the SSH model in the classification according to Bo-Hung [CITE: Two-Dimensional Extended Su-Schrieffer-Heeger Model; Masters Thesis; Chen, Bo-Hung; July 2018]. The basis for the lattice is given by the vectors b_0 and b_1.

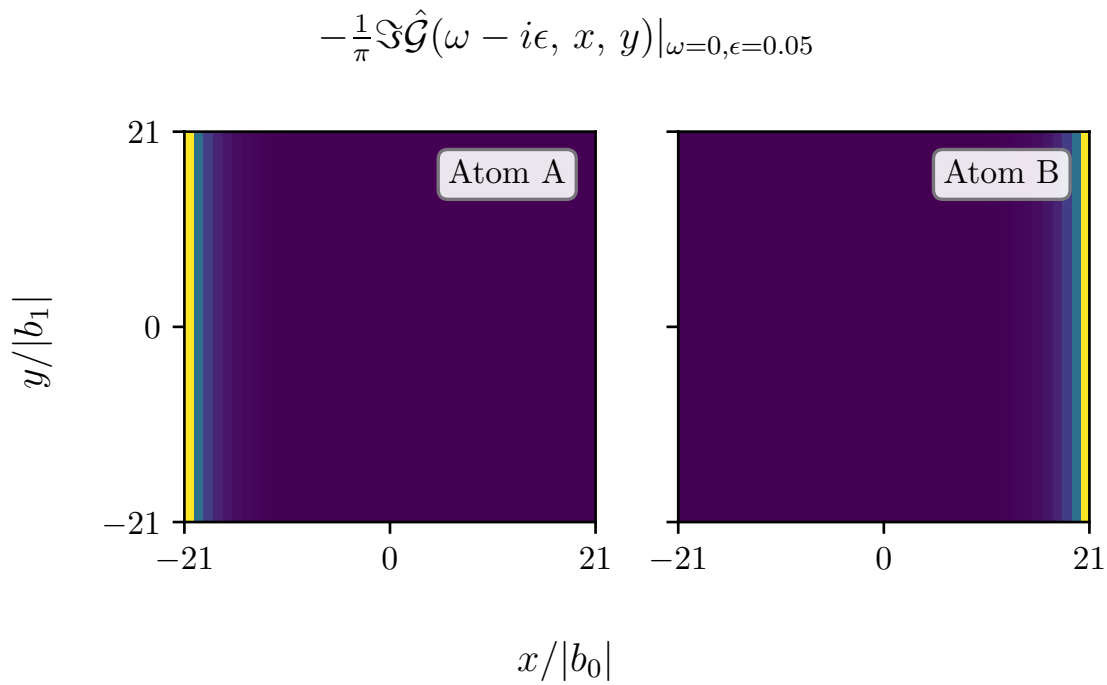

Figure 2: The local density of states of a Weyl SSH model, that is, a spinless square lattice with 43×43 lattice cells, with chemical potential $\mu/t = 0.00$, zero shift $s = 0$ and applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.05$. The SSH chains are in their topological phase $w = 1.2 > v = 0.6$, with diagonal hopping $t_d = 0.9$, which does gives rise to no topological modes perpendicular to the chains. Notice the zero bias density at the left edge for atom A and on the right for atom B, a signature of topological modes.

where $\psi_n(x, y)$ is the eigenvector associated with the nth eigenenergy, resolved at the point on the lattice (x, y). The Green's function matrix has a component for each atom. The Green's function is experimental observables through the *local density of states*,

$$\text{LDOS}(\omega; x, y) \equiv -\frac{1}{\pi}\Im\hat{\mathcal{G}}(\omega - i\epsilon, x, y), \tag{15}$$

which is a measure of the density of states at a given point, at a specified applied bias energy ω and resolution ϵ. The local density of states for each atomic site with the unit cells are given in Figure 2, showing the existence of zero energy edge states.

Resolving the local density states at the level of the unit cell (summing over the atoms) and integrating over the y-axis, we study the *spectrum* of the system. The spectrum shows edge modes present at zero energy at the edges, which are not present in the bulk. These

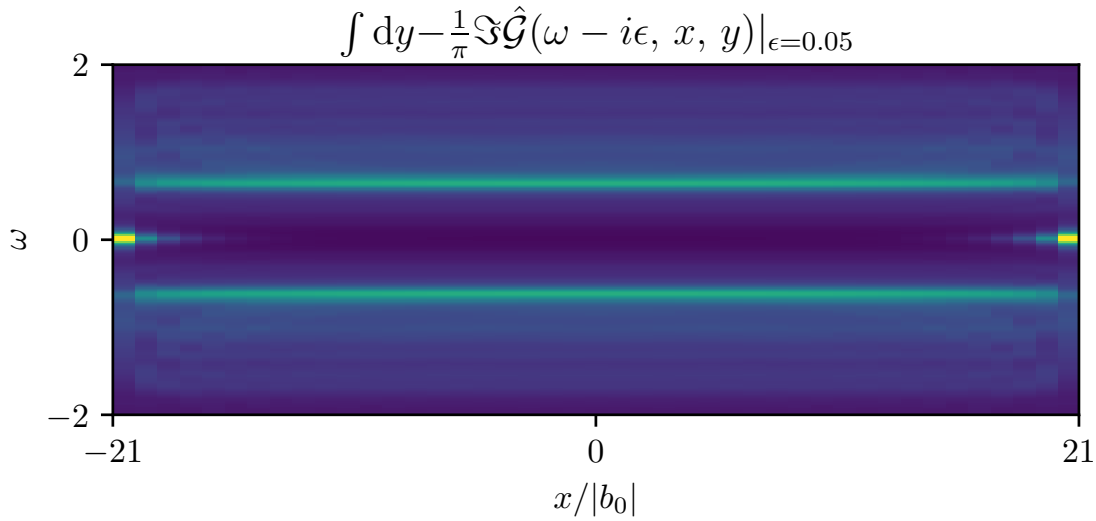

Figure 3: The spectrum of the Weyl-SSH model, in its normal state, integrated over the direction perpendicular to the SSH chains and summed over the atomic sites within the lattice cell. Zero-energy modes are observed at the edges, with a faint tail of density.

are distinct signatures of Majorana modes. In addition to localised density at the edge, there exists a 'tail' of density which the authors investigate further.

Majorana modes exist pinned to *Dirac points*, which are the cross point of two bands. Studying the band structure in Figure 11, we observe a zero-mode between the Dirac points, which is known as a *Bogoliubov-Fermi arc*.

In order to determine whether the modes are topological modes, it must be shown that they their wavefunctions are distributed nonlocally across the system. We plot the wavefunction via the related *spectral density* of the Green's function at the point k_y at which the Bogoliubov-Fermi arc and Majorana modes are are located, Figure 9. As a function of x-position, we observe that indeed, the modes are spread non-locally and that the arc tails.

We note that other hopping configurations are possible, leading either to topological modes along one pair of parallel edges, or along all edges; see [Two-Dimensional Extended Su-Schrieffer-Heeger Model; Bo Hung; Masters Thesis; National Taiwan Normal University] .

8 Results with impurities

The impurity is located at the centre with coupling strength $V = 1.21$.

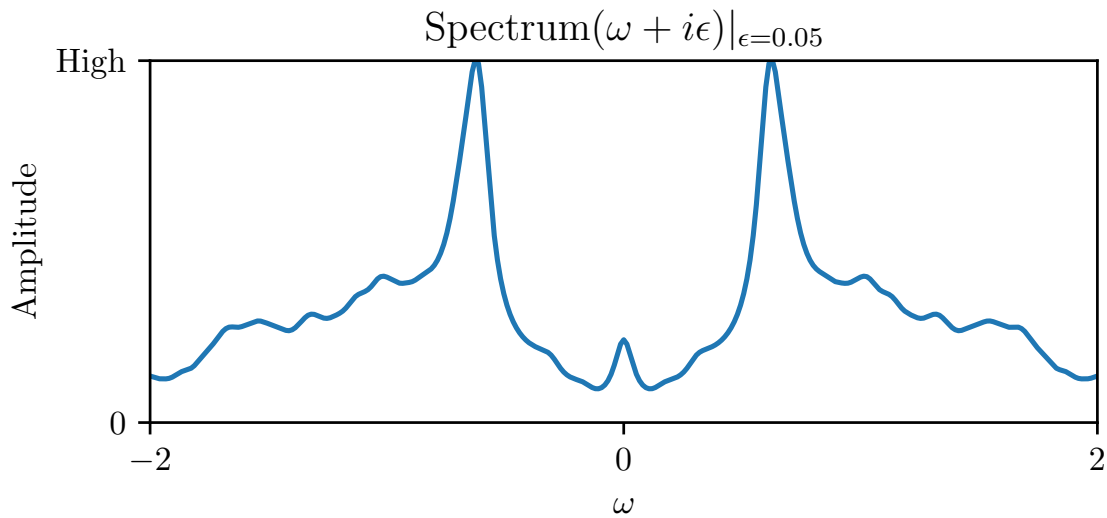

Figure 4: The spectrum of the Weyl-SSH model in the normal state with open-boundary conditions, summed over the atomic sites. The parameters are $\mu = 0.0, v = 0.6, w = 1.2$ and $t_d = 0.9$. We observe the topological mode as a density peak within the band gap.

The x-axis is glued to periodic boundary conditions.

Our investigation shows no interesting interplay between a single impurity and topological modes, however, impurities can be used to generate quasiparticle-interference patterns as an experimental diagnosis of topological modes. Moreover, a wall of impurities

9 Quasiparticle-interference patterns for Majorana and Bogoliubov-Fermi arcs

10 Phase diagram

11 Wall impurities

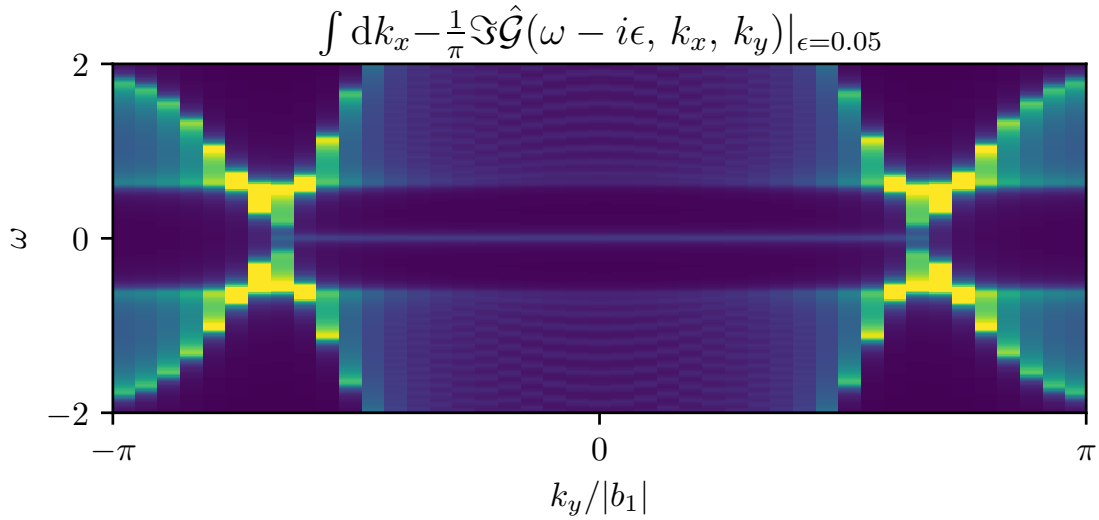

Figure 5: The spectrum of the Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. Dirac nodes are observed, which pin Majorana zero-modes. Between the Dirac nodes is a line of zero-energy states, known as a Bogoliubov-Fermi arc.

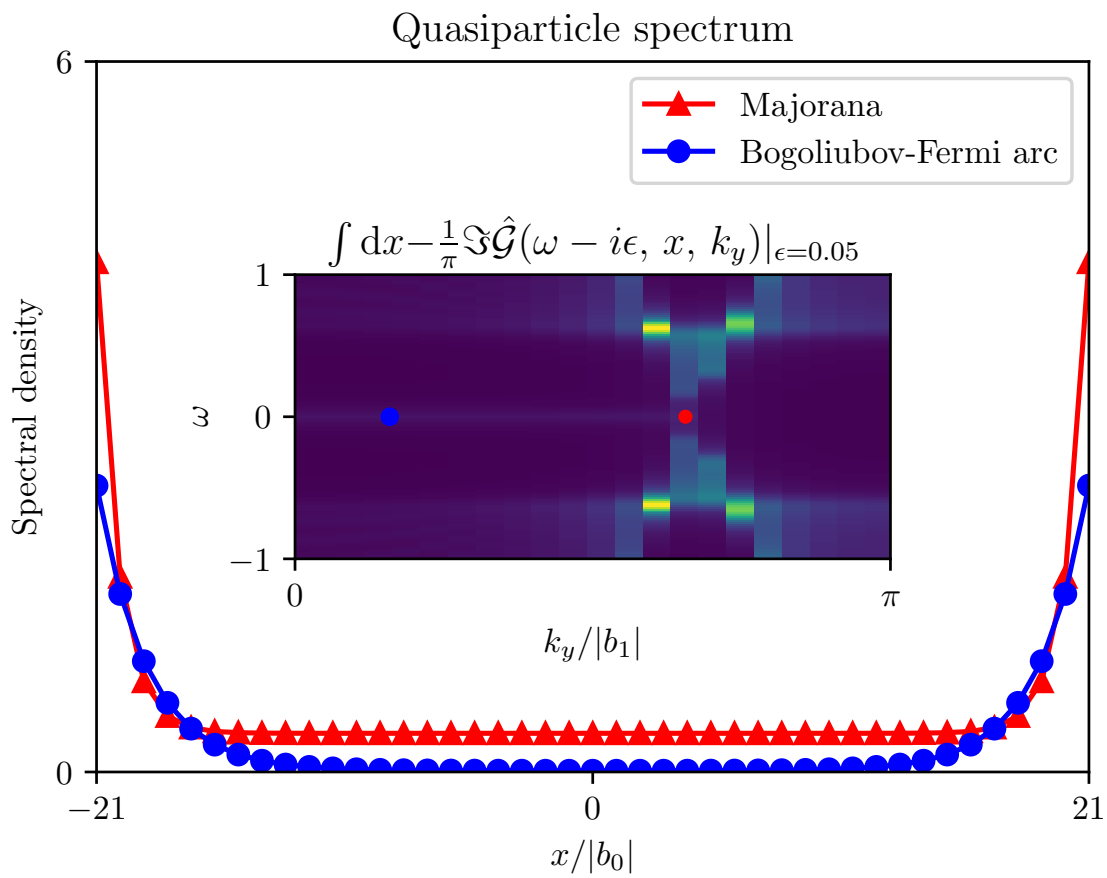

Figure 6: The spectrum of the Majorana and Bogoliubov-Fermi arc quasiparticle modes, at zero-applied bias, with the vertical momentum specified by the (colour-coordinated) spectral plot. The quasiparticle density is plotted over the direction parallel to the SSH chains. The Majorana mode (red) is localised at the edges. The Bogoliubov-Fermi arc, in contrast, tails.

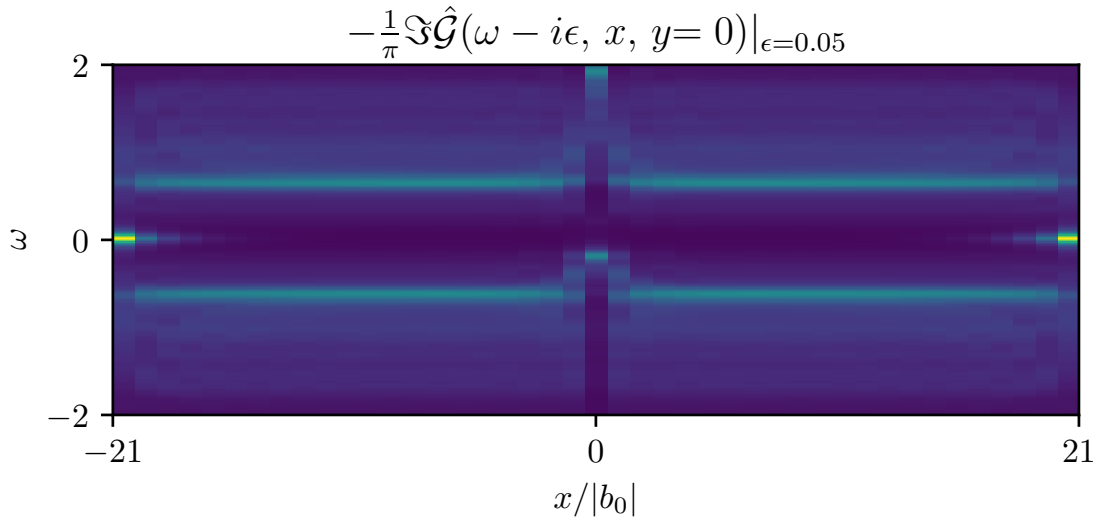

Figure 7: The local density of states along a line parallel to the axis of the dimers through a central impurity. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions along the axis. The impurity has coupling strength $V/t = 1.21$. No additional zero-mode is seen.

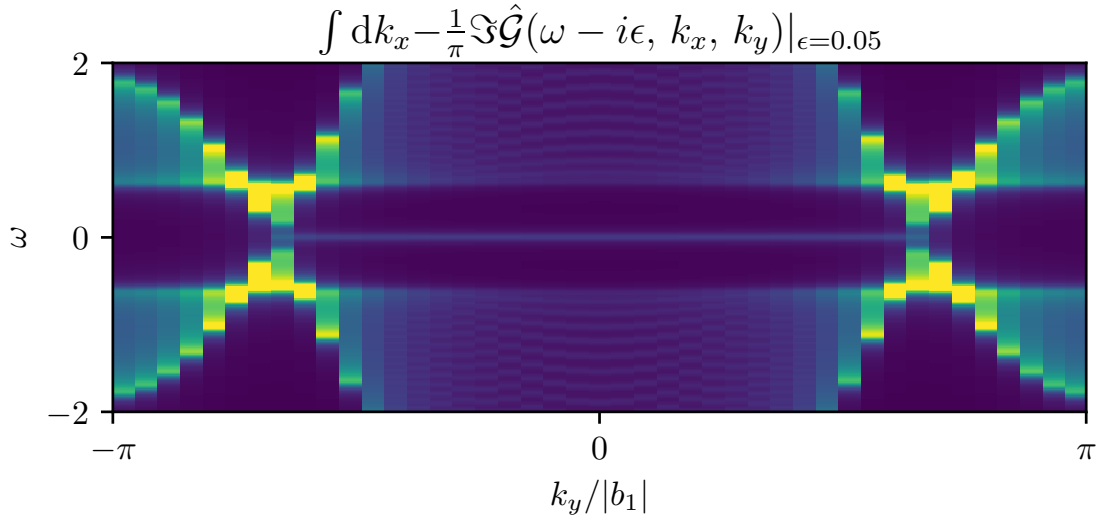

Figure 8: Band structure integrated over the axis of the SSH dimers, k_x, with open boundary conditions along the axis perpendicular to the dimers. No qualitative change in the spectrum is observed from the impurity-free model.

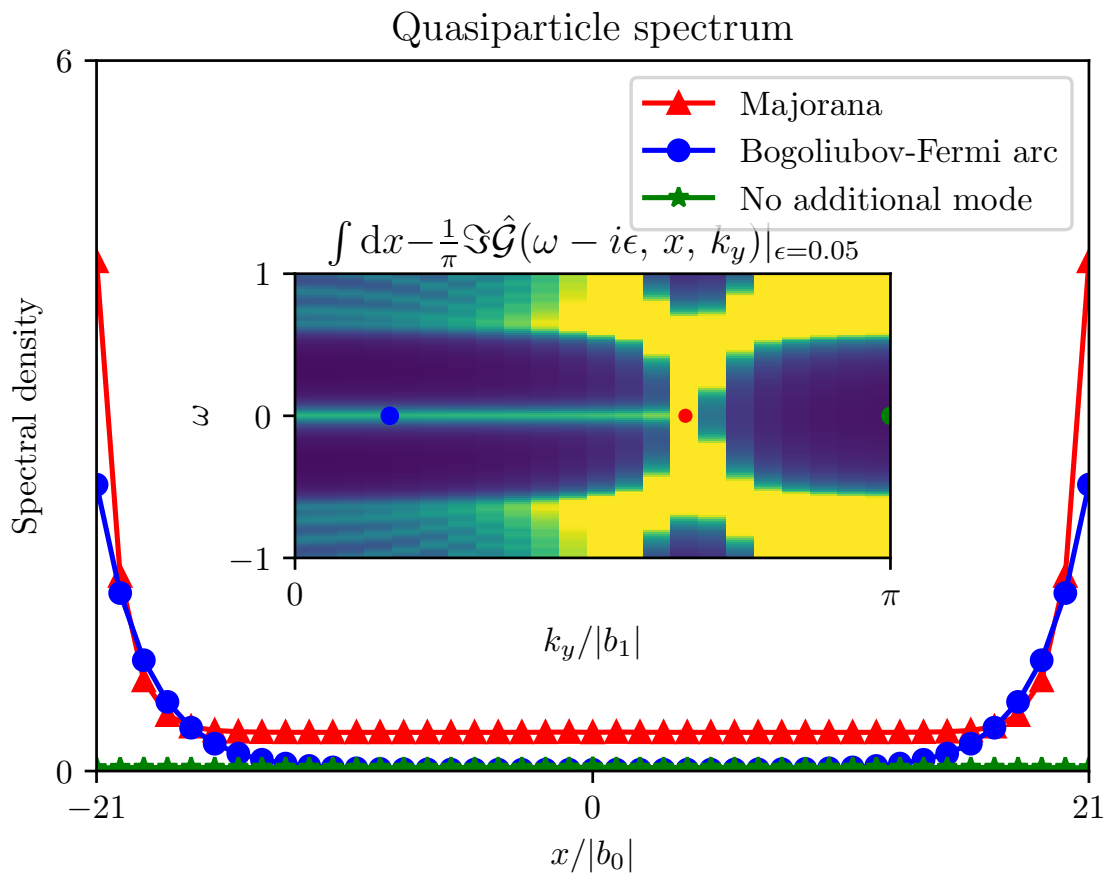

Figure 9: Topological modes in the presence of a single impurity. No localisation of the quasiparticles around the impurity are seen. Compare with Figure 12 –we observe an additional Bogoliubov-Fermi arc in a model with a wall of impurities. Note that the axis perpendicular to the dimers has been integrated over in the spectral density plot.

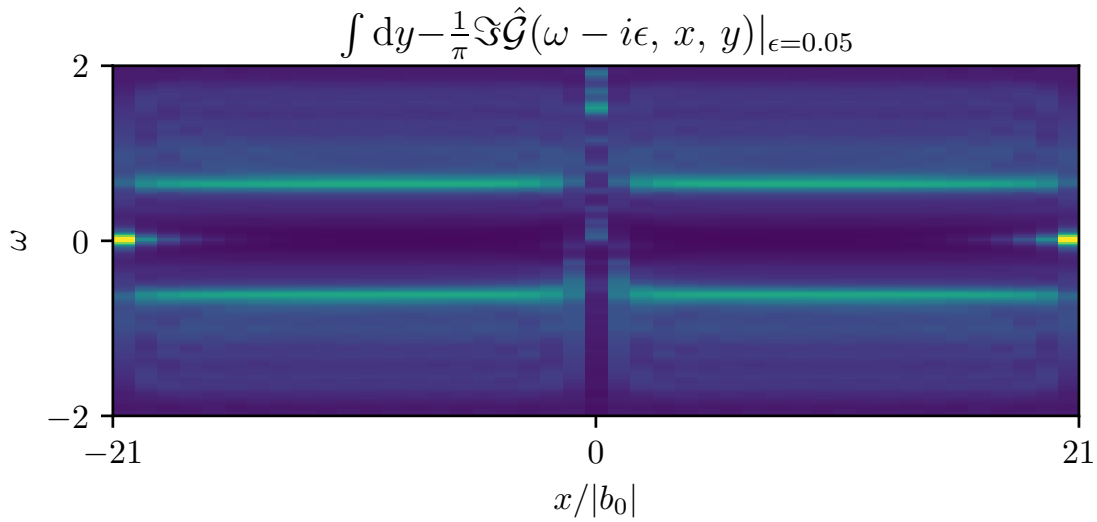

Figure 10: Local density of states with an impurity at the centre. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions perpendicular to the axis of the dimers, and open boundary conditions along the axis parallel. The impurity wall runs perpendicular to the axis and through the middle, with coupling strength $V/t = 1.21$. Additional modes are seen although at such weak coupling, it is not clear whether they are gapped.

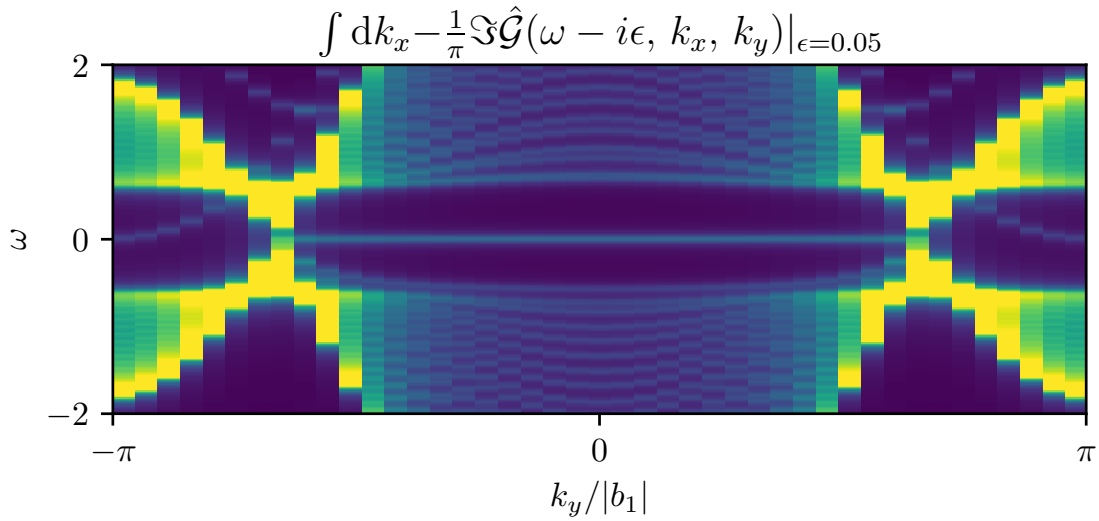

Figure 11: Local density of states integrated over the axis of the SSH dimers and summed over the intracell atomic sites show clearly that there exists an additional gapped mode due to the impurity wall. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions parallel. The impurity wall has coupling strength $V/t = 1.21$.

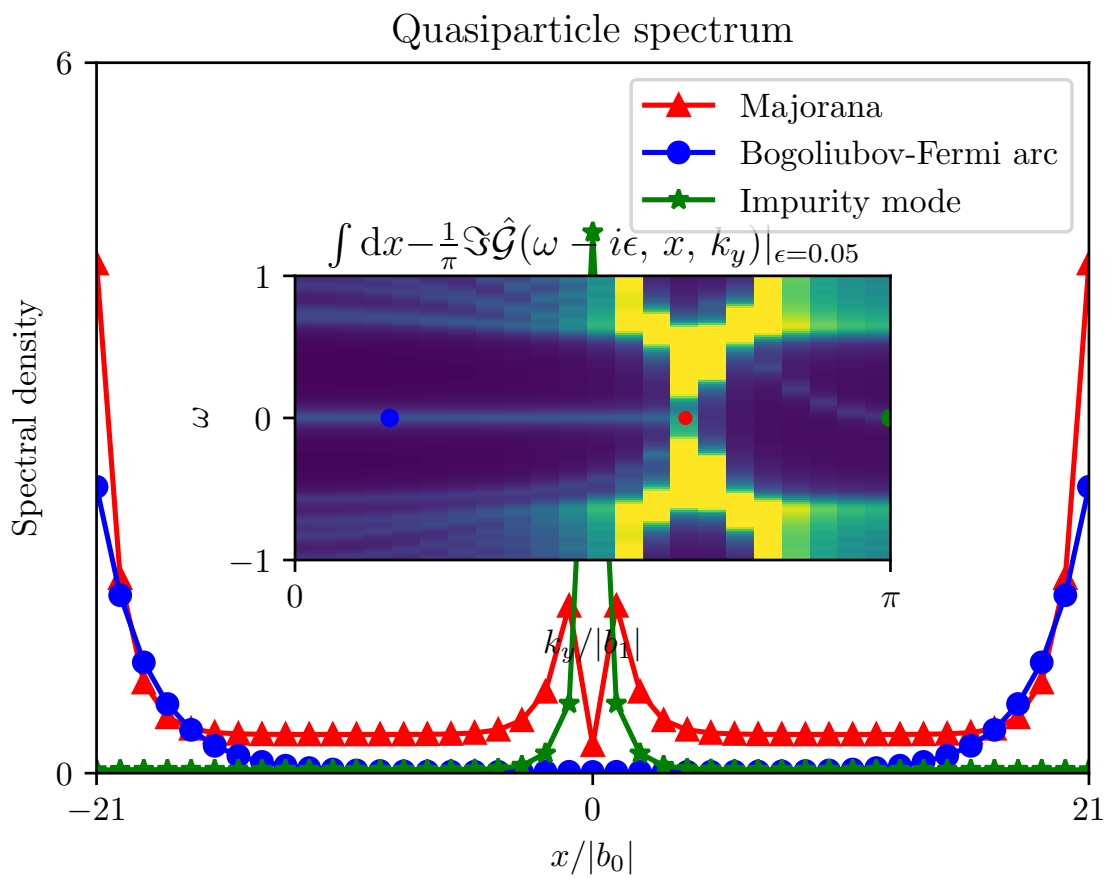

Figure 12: The majorana quasiparticle exhibits some pinning around the impurity wall and an additional arc mode exists at $k_y = \pi$, whose nature is to be investigated.

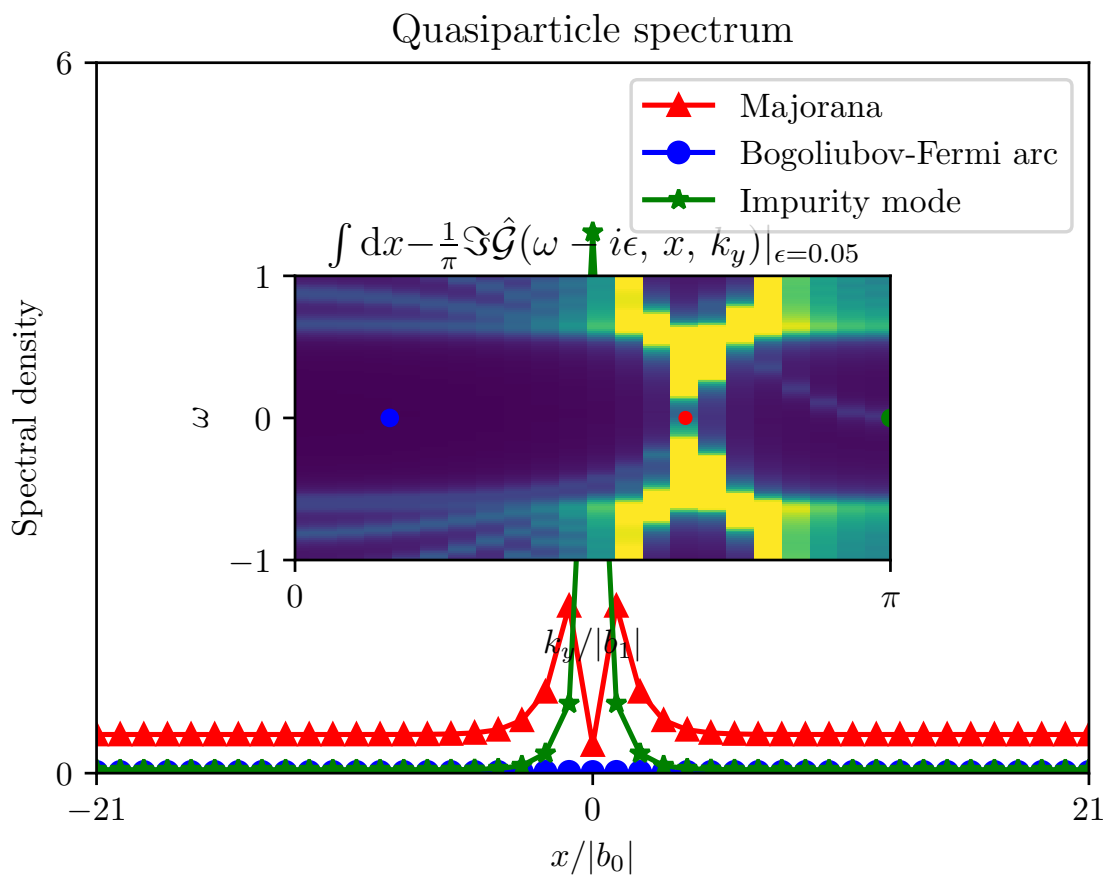

Figure 13: Investigation of the modes with periodic boundary conditions along both axes. The additional mode is shown to be due to the presence of a wall of impurities.

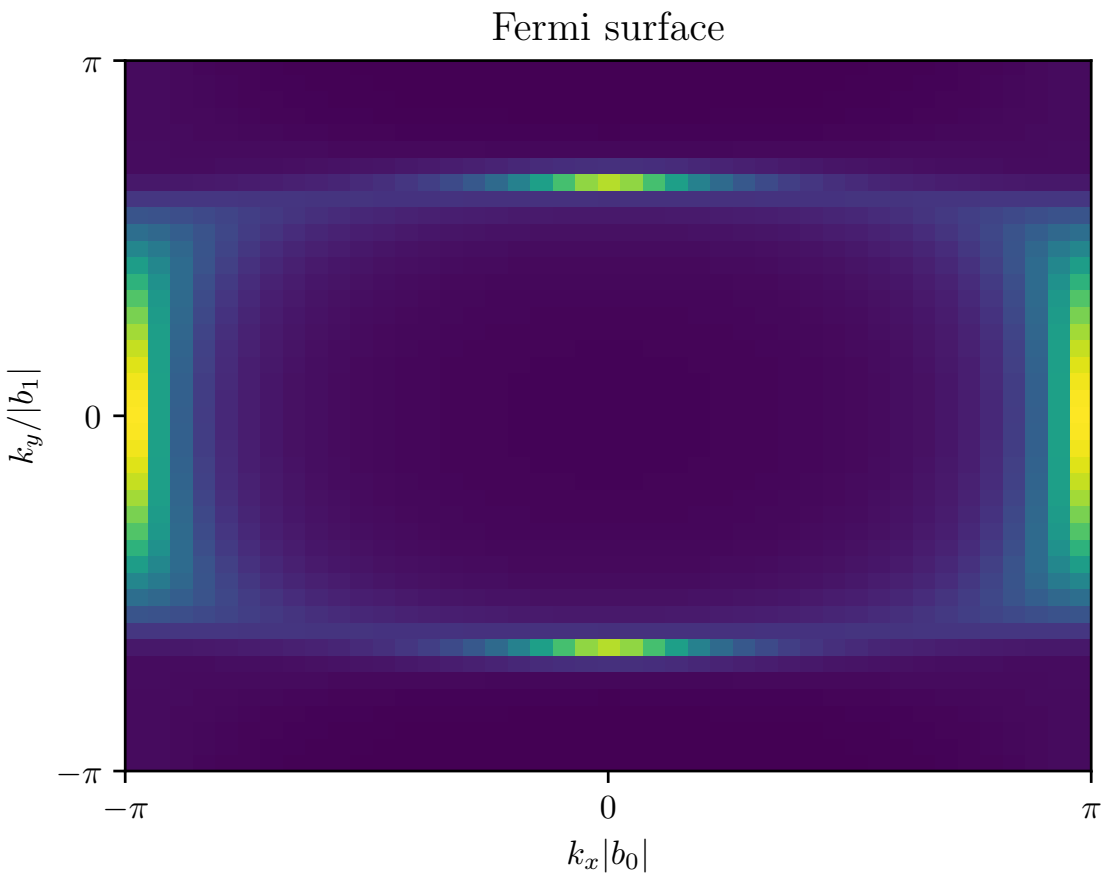

Figure 14: Fermi surface with a central impurity with coupling strength $V = 1.21$ and chemical potential $\mu = 0.0$, in the normal state. Notice density concentrated near $k_y \approx \pi/2$ corresponding to the Majorana mode, and density around $k_x \approx \pm\pi$ corresponding to the Fermi arc.

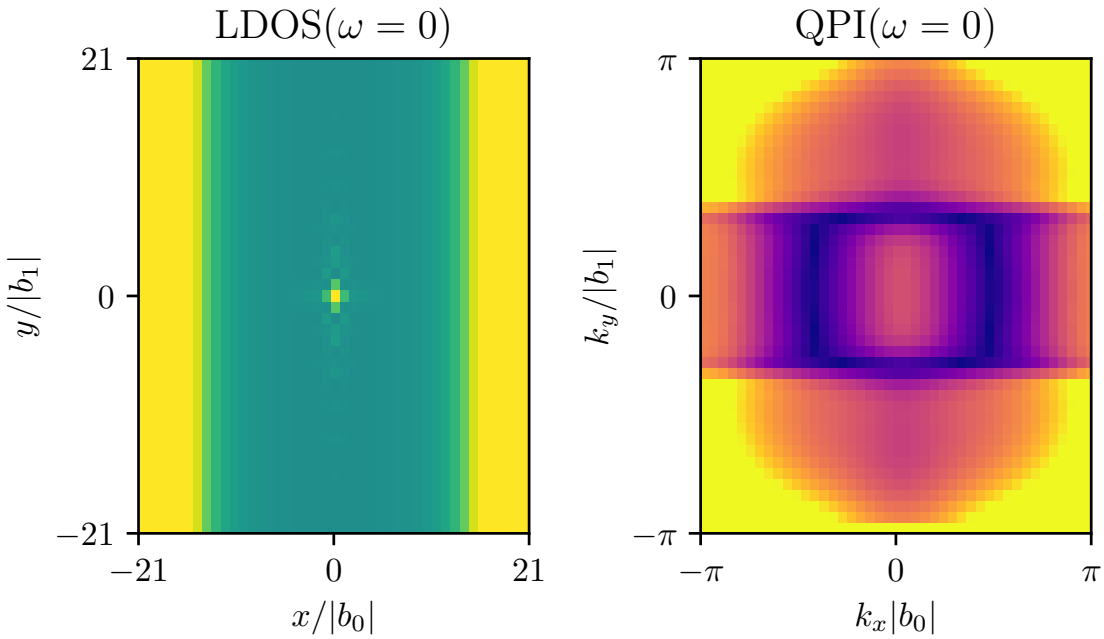

Figure 15: The local density of states and quasiparticle interference pattern for an 43×43 cell Weyl-SSH model with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal, with a central impurity with coupling strength $V/t = 1.21$. The chemical potential is $\mu/t = 0.00$.

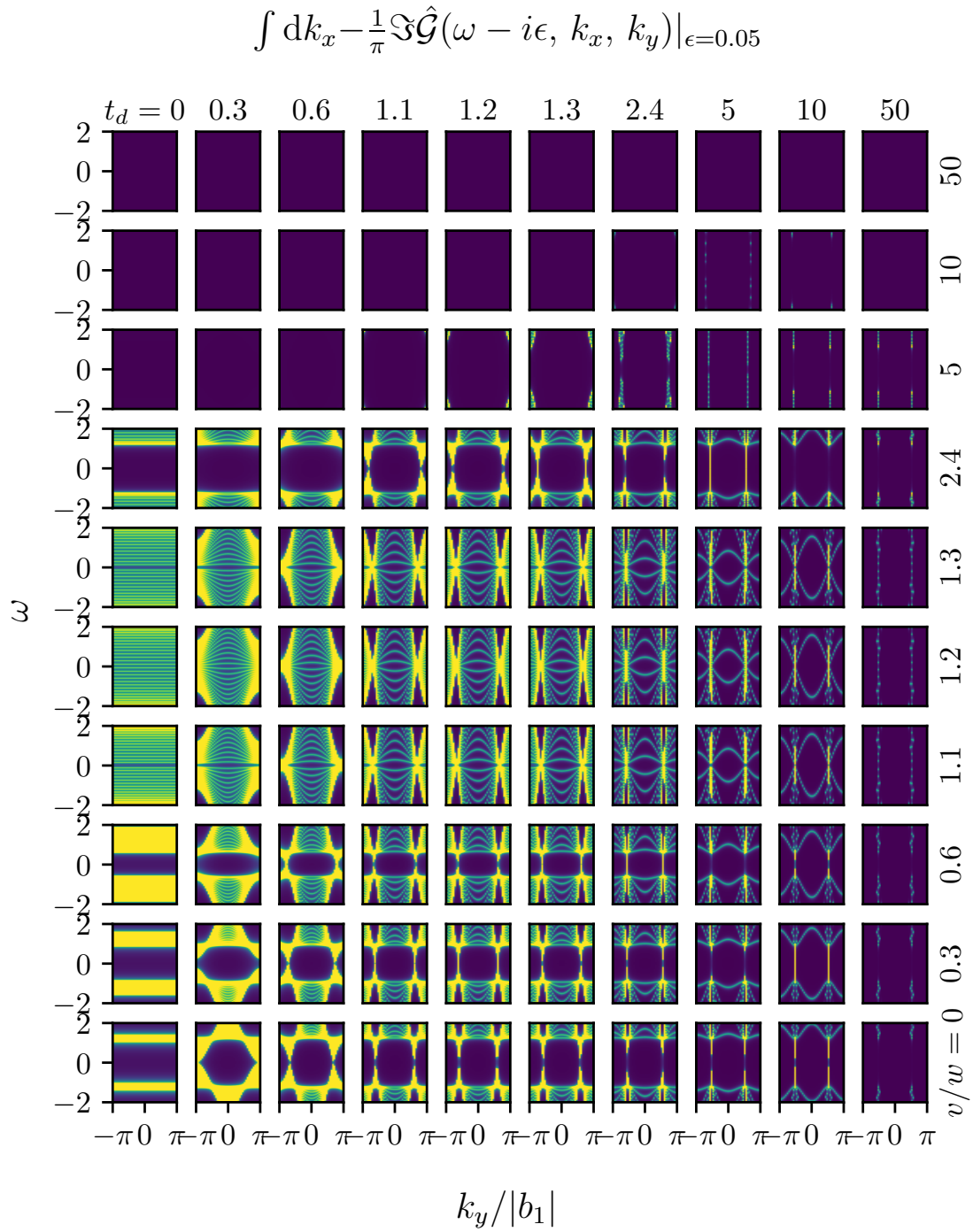

Figure 16: The spectrum of the simple Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The model consists of an 43×43 lattice, with closed boundary conditions. The chemical potential is $\mu = 0.0$, the interdimer hopping is fixed $w = 1.2$, and the intradimer v and interchain t_d hoppings are varied. Compare with the model with open boundary conditions Fig **??**, and notice the universality when comparing to the NiC_2 model Fig **??**, **??**.

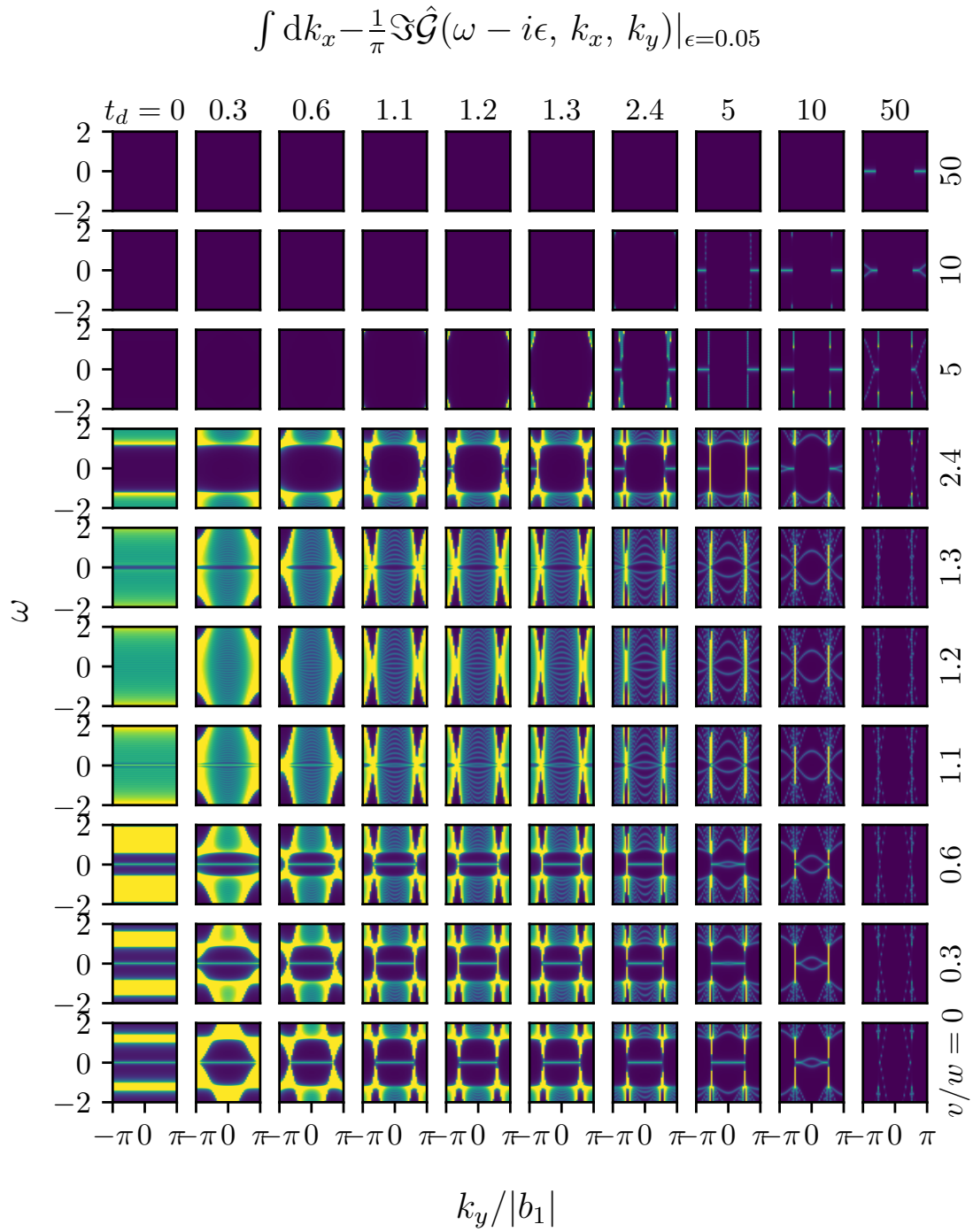

Figure 17: The model and quantities depicted as in Fig 16, but with the boundary conditions open. We observe the diagonal hopping t_d gives rise to two Dirac cones. In the SSH topological phase, a Bogliubov-Fermi arc connects them through the centre of the Brillouin zone. Then, in the SSH trivial phase, we find the Fermi arc instead passes through the edge of the Brillouin zone.

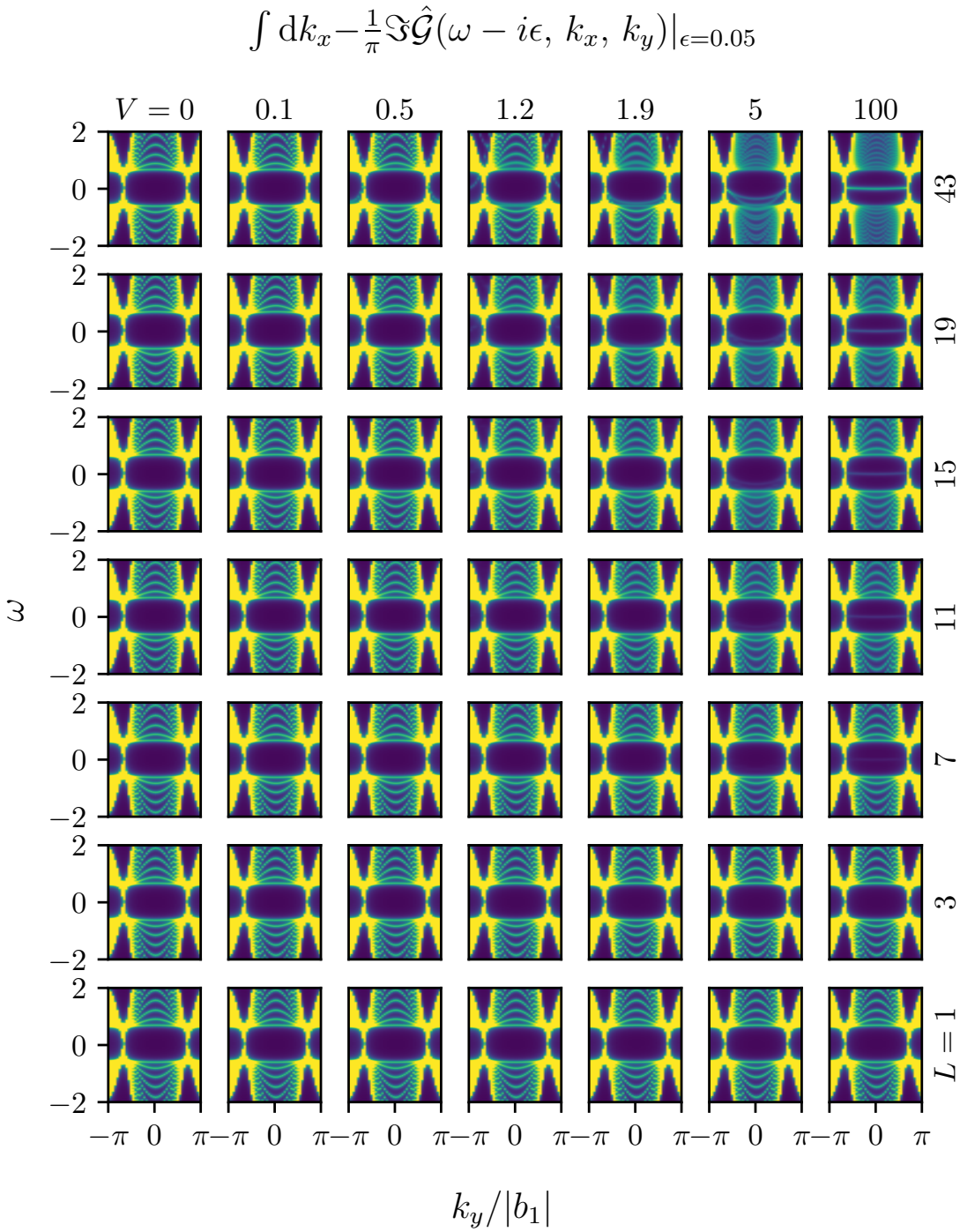

Figure 18: Phase diagram of an impurity wall, perpendicular to the SSH dimers, of length L and coupling strength V, and with closed boundary conditions. The spectrum of the simple Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The model consists of an 43×43 lattice, with chemical potential $\mu = 0.0$, interdimer $w = 1.2$, intradimer $v = 0.6$ and interchain $t_d = 0.9$ hoppings. Compare with the model with open boundary conditions Fig 19.

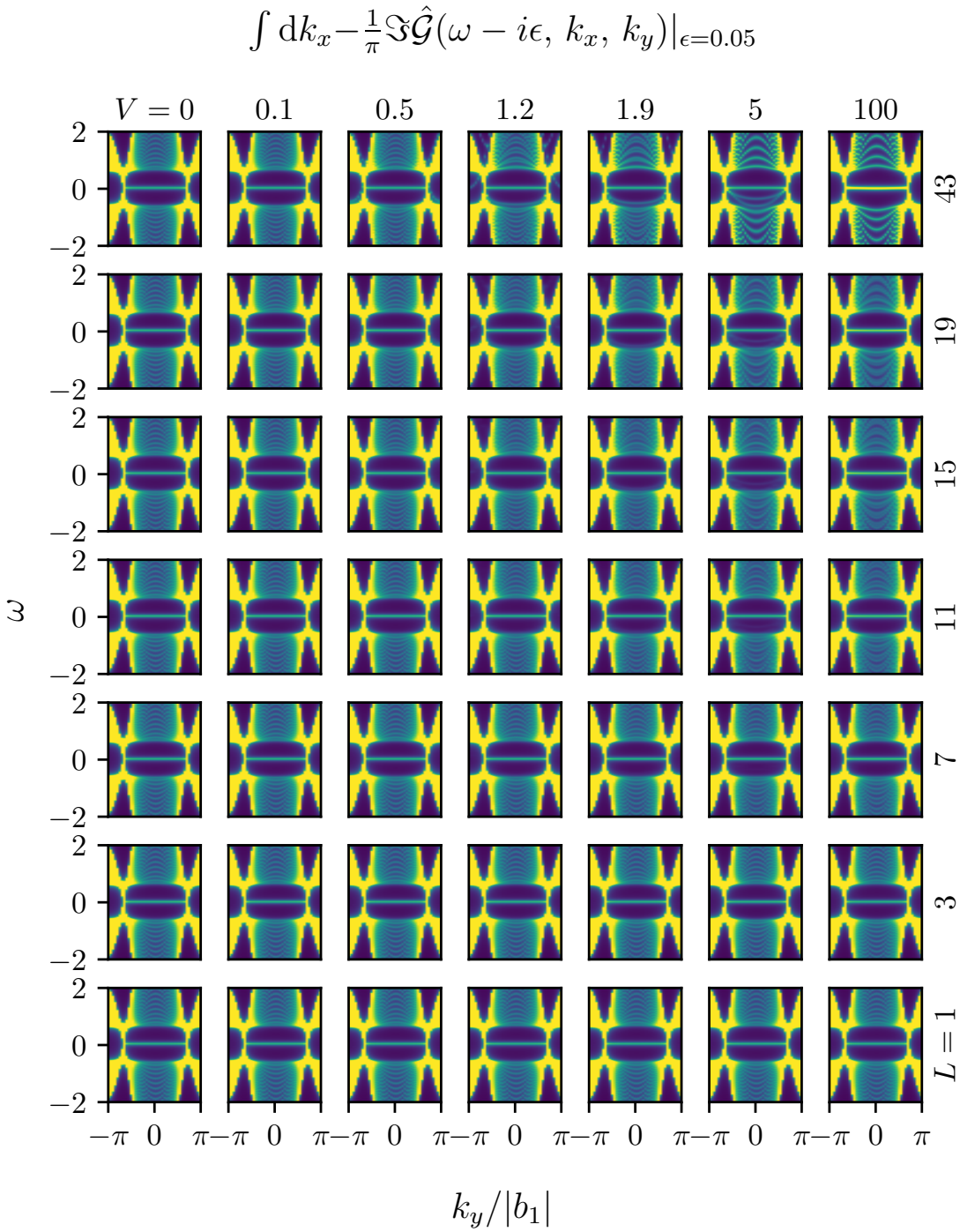

Figure 19: The model and quantities depicted as in Fig 18, but with the boundary conditions open. We observe the topological mode due to the impurity is independent of the topological modees assoicated with the boundary conditions.

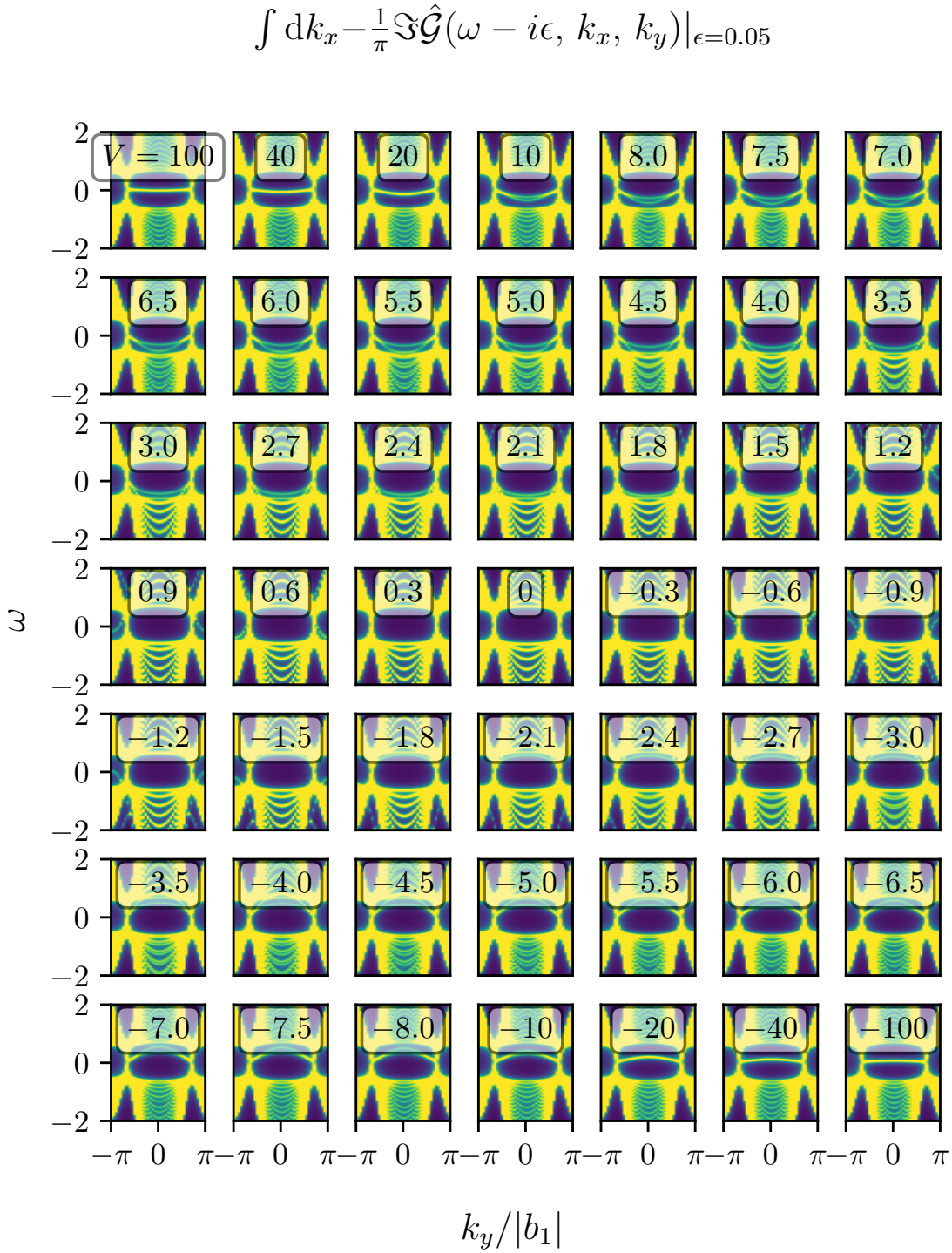

Figure 20: Impurity wall. The spectrum of the Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites, 43×43 cells, with closed boundary conditions. The chemical potential is $\mu = 0.0$, interdimer hopping $w = 1.2$, intradimer $v = 0.6$ and interchain $t_d = 0.9$, corresponding to the topological phase of the SSH model. The potential barrier is attractive with negative $V < 0$ and repulsive with positive $V > 0$. For small $|V| \approx w$, a mode outside the Brillouin Zone leaves the bulk. In particular, zero-energy modes exist at the edge of the Brillouin zone when $|V| = w$. As $|V| > w$ increases, the mode moves into the centre of the Brillouin Zone, falling or rising with the sign of V, tending to the topological spectrum of the open-

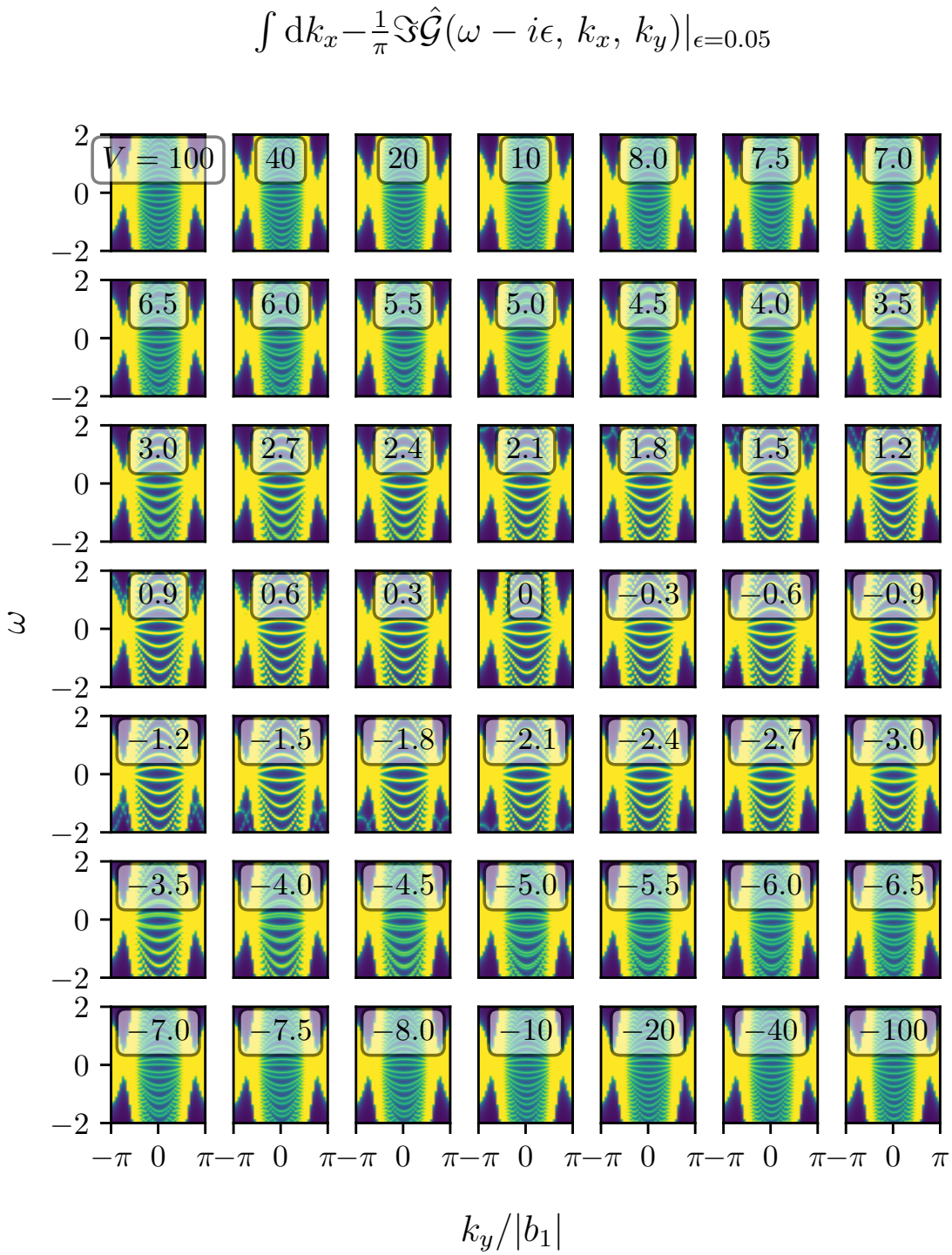

Figure 21: Model and quantities depicted in Figure 20, except with the interdimer hopping $v = w = 1.2$, the topological transition point of the SSH model.

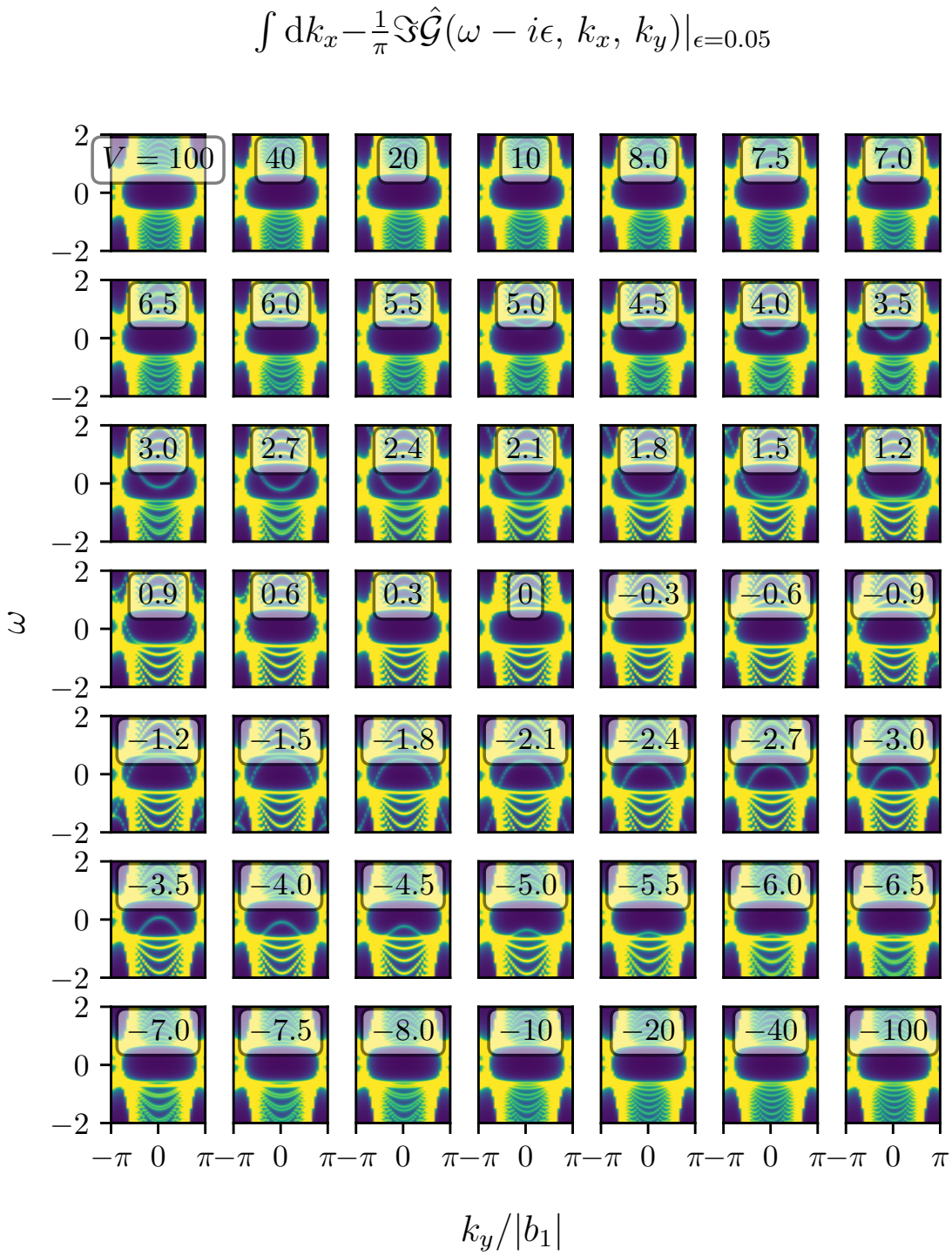

Figure 22: Model and quantities depicted in Figure 20, except with the interdimer hopping $v = 1.8 > w$, the trivial phase of the SSH model.